

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408257
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Onyshchenko; Yuliia et al.

Device and method for generating a plasma jet

Abstract

A device for generating a plasma jet is disclosed. A nozzle of the device comprises an inner, middle and outer channel surrounding each other and electrically insulated from each other. The device is configured for flowing through the inner, middle and outer channel respectively a first ignitable gas, a monomer precursor and a second ignitable gas towards their respective exit openings. A high voltage electrode is arranged in the inner channel for generating a plasma from the first ignitable gas. The nozzle further comprises a nozzle cap enclosing a mixing volume after the exit openings of the inner, middle and outer channel. The nozzle cap is provided with an exit opening after the mixing volume, which exit opening is smaller than the exit opening of the outer channel.

Inventors:	Onyshchenko; Yuliia (Zwijnaarde, BE), Narimisa; Mehrnoush (Heusden, BE), De Geyter; Nathalie (Langemark, BE), Morent; Rino (Langemark, BE)
Applicant:	UNIVERSITEIT GENT (Ghent, BE)
Family ID:	75377648
Assignee:	UNIVERSITEIT GENT (Ghent, BE)
Appl. No.:	18/553715
Filed (or PCT Filed):	March 28, 2022
PCT No.:	PCT/EP2022/058144
PCT Pub. No.:	WO2022/207563
PCT Pub. Date:	October 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240188208 A1	Jun. 06, 2024

Foreign Application Priority Data

EP21166684Apr. 01, 2021

Publication Classification

Int. Cl.: H05H1/46 (20060101); H01J37/32 (20060101); B05D1/00 (20060101)

U.S. Cl.:

CPC H05H1/466 (20210501); H01J37/32091 (20130101); H01J37/3244 (20130101); B05D1/62 (20130101); H01J2237/3382 (20130101)

Field of Classification Search

CPC: H05H (1/466); H01J (37/32091); H01J (37/3244); H01J (2237/3382); B05D (1/62)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8883027	12/2013	Yoon	156/345.43	H01L 21/67069
8926789	12/2014	Yoon	156/345.31	H01L 21/02087
11810756	12/2022	Yancey	N/A	B44D 3/16
2011/0121107	12/2010	Siffer et al.	N/A	N/A
2012/0100300	12/2011	Gindrat	427/446	H05H 1/3405
2014/0120731	12/2013	Xu et al.	N/A	N/A
2017/0243727	12/2016	Poenitzsch	N/A	B05B 7/224
2017/0338084	12/2016	Nishijima	N/A	H01J 37/3255
2018/0047595	12/2017	Kofuji	N/A	H01J 37/32357
2020/0098651	12/2019	Sakiyama	N/A	C23C 16/50
2022/0339373	12/2021	De Geyter	N/A	H05H 1/2443
2024/0188208	12/2023	Onyshchenko	N/A	H01J 37/32091

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2320715	12/2010	EP	N/A
3586954	12/2022	EP	N/A
20110049983	12/2010	KR	N/A
2009036579	12/2008	WO	N/A
WO-2014191012	12/2013	WO	H05H 1/2406

OTHER PUBLICATIONS

European Search Report and Written Opinion for EP21166684.7 dated Sep. 15, 2021, 8 pages. cited by applicant

International Search Report and Written Opinion for PCT/EP2022/058144 dated Jul. 26, 2022, 12 pages. cited by applicant

Primary Examiner: Houston; Adam D

Attorney, Agent or Firm: Dickinson Wright PLLC

Background/Summary**CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) This is a National Stage Application, filed under 35 U.S.C. 371 based on International Patent Application No. PCT/EP2022/058144, filed Mar. 28, 2022, which claims priority to EP Patent Application Ser. No. 21/166,684.7 filed Apr. 1, 2021, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a device and a method for generating a plasma jet, preferably for plasma polymerisation or plasma activation.

BACKGROUND ART

(3) Devices for generating a plasma jet are known from the prior art, for example from WO 2018/141802 A1 or EP 3 586 954 A1. Such a device comprises a nozzle through which plasma is ejected in the form of a jet. The nozzle comprises a first channel, and the device is configured for flowing an ignitable gas, suitable to generate a plasma therefrom, through the first channel towards an exit opening of the nozzle. A high voltage electrode is arranged in the first channel for generating a plasma from the ignitable gas flowing in the first channel. The device is further provided for mixing a monomer precursor with the ignitable gas or the plasma generated therefrom in the first channel, such that the monomer precursor can be polymerised in the generated plasma. The monomer precursor can be provided from a second channel into the first channel, but can also be provided directly into the first channel together with the ignitable gas.

(4) Such devices for generating a plasma jet known from the prior art have the disadvantage that the monomer precursor is often not well mixed with the ignitable gas or the plasma generated therefrom, such that not all monomer precursor is effectively polymerised in the plasma jet.

DISCLOSURE OF THE INVENTION

(5) It is an aim of the present invention to provide a device and a method for generating a plasma jet having a better mixing of a monomer precursor in the plasma jet.

(6) This aim is achieved according to the present invention with a device and a method for generating a plasma jet showing the technical characteristics of respectively the first independent claim and the second independent claim.

(7) Therefore, the present invention provides a device for generating a plasma jet. Preferably, the plasma jet comprises a monomer precursor. Preferably; the plasma jet is for plasma polymerisation or plasma activation, but not necessarily limited thereto. The device comprises a nozzle. The nozzle of the device comprises an inner channel. The device is configured for flowing a first ignitable gas through the inner channel towards an exit opening of the inner channel. A high voltage electrode is arranged in the inner channel for generating a plasma from the first ignitable gas flowing in the

inner channel. Preferably, the high voltage electrode is arranged along the length of the inner channel. Preferably, the high voltage electrode extends up to the exit opening of the inner channel. Preferably, a pointed tip of the high voltage electrode is located at the exit opening of the inner channel. The nozzle of the device comprises a middle channel. The middle channel surrounds the inner channel. The middle channel is electrically insulated from the inner channel. The device is configured for flowing a monomer precursor through the middle channel towards an exit opening of the middle channel. The nozzle of the device comprises an outer channel. The outer channel surrounds the middle channel. The outer channel is electrically insulated from the middle channel. The device is configured for flowing a second ignitable gas through the outer channel towards an exit opening of the outer channel. Preferably, the inner channel, the middle channel and the outer channel are circular channels. Preferably, the inner channel, the middle channel and the outer channel are coaxial channels. Preferably, the exit openings of the inner channel, the middle channel and the outer channel are coplanar. The nozzle of the device comprises a nozzle cap. The nozzle cap encloses a mixing volume located after the exit openings of the inner channel, the middle channel and the outer channel. Preferably, the nozzle cap is surrounding an end portion of the outer channel, the middle channel and the inner channel. Preferably, the nozzle cap is rotationally symmetric. Preferably, the nozzle cap is coaxial with the coaxial inner channel, middle channel and outer channel. Preferably, the nozzle cap is in an electrically conducting material. Preferably, the nozzle cap is grounded. The nozzle cap is provided with an exit opening after the mixing volume. The exit opening of the nozzle cap is smaller than the exit opening of the outer channel. Preferably, the exit opening of the nozzle cap is circular. Preferably, the exit opening of the nozzle cap is arranged in front of at least the exit opening of the inner channel. Preferably, the exit opening of the nozzle cap is coaxial with the coaxial inner channel, middle channel and outer channel. Preferably, the exit opening of the nozzle cap is arranged in a plane parallel to the plane of the exit openings of the inner channel, the middle channel and the outer channel.

(8) The nozzle cap enclosing a mixing volume after the exit openings of the inner channel, the middle channel and the outer channel, and the nozzle cap having an exit opening after the mixing volume with a size smaller than the exit opening of the outer channel, offers the advantage that the combined flow of the plasma and possible leftover first ignitable gas, the monomer precursor, and the second ignitable gas out of the exit openings of respectively the inner channel, the middle channel and the outer channel cannot pass completely through the smaller exit opening of the nozzle cap and is circulated inside the mixing volume. With this combined flow being circulated inside the mixing volume a better mixing of the monomer precursor with the plasma and/or the ignitable gases, i.e. the first ignitable gas and the second ignitable gas, is achieved. Thereby, further plasma can be generated from the ignitable gases circulated in the mixing volume by means of the high voltage electrode in the inner channel, which preferably extends up to the exit opening of the inner channel and thus up to the mixing volume.

(9) The outer channel providing a flow of the second ignitable gas in the mixing volume, and the exit opening of the nozzle cap being smaller than the exit opening of the outer channel, offers the advantage that the flow of the second ignitable gas is directed towards the flow of the plasma and possible leftover first ignitable gas from the inner channel and the flow of the monomer precursor from the middle channel, thereby preventing these flows from directly exiting the mixing volume through the exit opening of the nozzle cap, and thereby aiding in circulating the different flows inside the mixing volume of the nozzle cap to improve the mixing of the monomer precursor with the plasma and/or the ignitable gases.

(10) The outer channel providing a flow of the second ignitable gas in the mixing volume, and the exit opening of the nozzle cap being smaller than the exit opening of the outer channel, also offers the advantage that the flow of the second ignitable gas is guided along an inner wall of the nozzle cap enclosing the mixing volume towards the exit opening of the nozzle cap. This flow of the second ignitable gas shields the inner wall of the nozzle cap enclosing the mixing volume from

polymerised monomer precursor in the mixing volume being deposited on said inner wall, which is undesirable since it would clog the nozzle.

(11) In an embodiment of the device according to the present invention the mixing volume is adapted for mixing the plasma of the first ignitable gas from the inner channel, the monomer precursor from the middle channel and the second ignitable gas from the outer channel.

(12) In an embodiment of the device according to the present invention the mixing volume is located within the nozzle cap.

(13) In an embodiment of the device according to the present invention the mixing volume is located directly after the exit openings of the inner channel, the middle channel and the outer channel.

(14) In an embodiment of the device according to the present invention the exit opening of the nozzle cap is located directly after the mixing volume.

(15) In an embodiment of the device according to the present invention the mixing volume extends from the exit openings of the inner channel, the middle channel and the outer channel to the exit opening of the nozzle cap.

(16) In an embodiment of the device according to the present invention the exit opening of the nozzle cap is smaller than or equal to the exit opening of the middle channel.

(17) This embodiment is beneficial for fully directing the flow of the second ignitable gas out of the exit opening of the outer channel towards the flow of the plasma and possible leftover first ignitable gas from the inner channel and the flow of the monomer precursor from the middle channel, thereby further preventing these flows from directly exiting the mixing volume through the exit opening of the nozzle cap, and thereby aiding in circulating the different flows inside the mixing volume of the nozzle cap to further improve the mixing of the monomer precursor with the plasma and/or the ignitable gases.

(18) In an embodiment of the device according to the present invention an inner wall of the nozzle cap is shaped for gradually guiding the flow of the second ignitable gas in the mixing volume from the exit opening of the outer channel towards the exit opening of the nozzle cap.

(19) The inner wall of the nozzle cap enclosing the mixing volume being shaped as such is beneficial for maintaining a stable flow of the second ignitable gas along the inner wall of the nozzle cap towards the exit opening of the nozzle cap, which improves the shielding by the second ignitable gas of the inner wall against polymerised monomer precursor in the mixing volume being deposited on said inner wall.

(20) In an embodiment of the device according to the present invention the inner wall of the nozzle cap is funnel shaped for gradually guiding the flow of the second ignitable gas in the mixing volume from the exit opening of the outer channel towards the exit opening of the nozzle cap.

(21) The inventors have found that the funnel shape of the inner wall of the nozzle cap is a very suitable shape for gradually guiding the flow of the second ignitable gas in the mixing volume from the exit opening of the outer channel towards the exit opening of the nozzle cap.

(22) In an embodiment of the device according to the present invention the nozzle cap is arranged for being removably connected to the nozzle of the device.

(23) This embodiment offers the advantage that the nozzle cap can be removed from the nozzle for cleaning, or for replacement with a different type of nozzle cap, for example a nozzle cap having an exit opening of a different size.

(24) In an embodiment of the device according to the present invention the device is configured for flowing the first ignitable gas at a predetermined first flow rate through the inner channel towards the exit opening of the inner channel. In an embodiment of the device according to the present invention the first flow rate is at least 0.1 slm, preferably at least 0.2 slm, more preferably at least 0.3 slm, and even more preferably at least 0.4 slm. In an embodiment of the device according to the present invention the first flow rate is at most 0.9 slm, preferably at most 0.8 slm, more preferably at most 0.7 slm, and even more preferably at most 0.6 slm. In an embodiment of the device

according to the present invention the first flow rate is most preferably 0.5 slm.

(25) The inventors have found that with the first flow rate of the first ignitable gas in the inner channel, and thus also of the plasma generated from the first ignitable and possible leftover first ignitable gas entering the mixing volume, being in the given range, a uniform mixing of the monomer precursor with the plasma and the ignitable gases inside the mixing volume is achieved.

(26) In an embodiment of the device according to the present invention the device is configured for flowing the monomer precursor at a predetermined second flow rate through the middle channel towards the exit opening of the middle channel. In an embodiment of the device according to the present invention the second flow rate is at least 50 sccm, preferably at least 100 sccm, more preferably at least 150 sccm, and even more preferably at least 200 sccm. In an embodiment of the device according to the present invention the second flow rate is at most 650 sccm, preferably at most 550 sccm, more preferably at most 450 sccm, and even more preferably at most 350 sccm. In an embodiment of the device according to the present invention the second flow rate is most preferably 250 sccm.

(27) The inventors have found that with the second flow rate of the monomer precursor in the middle channel, and thus also of the monomer precursor entering the mixing volume, being in the given range, a uniform mixing of the monomer precursor with the plasma and the ignitable gases inside the mixing volume is achieved.

(28) In an embodiment of the device according to the present invention the device is configured for flowing a carbon based monomer precursor through the middle channel towards the exit opening of the middle channel. The carbon based monomer precursor is for example styrene or methyl methacrylate, MMA.

(29) In an embodiment of the device according to the present invention the device is configured for flowing the second ignitable gas at a predetermined third flow rate through the outer channel towards the exit opening of the outer channel. In an embodiment of the device according to the present invention the third flow rate is at least 0.2 slm, preferably at least 0.4 slm, more preferably at least 0.6 slm, and even more preferably at least 0.8 slm. In an embodiment of the device according to the present invention the third flow rate is at most 5 slm, preferably at most 4 slm, more preferably at most 3 slm, and even more preferably at most 2 slm. In an embodiment of the device according to the present invention the third flow rate is most preferably 1 slm.

(30) The inventors have found that with the third flow rate of the second ignitable gas in the outer channel, and thus also of the second ignitable gas entering the mixing volume, being in the given range, a uniform mixing of the monomer precursor with the plasma and the ignitable gases inside the mixing volume is achieved.

(31) In an embodiment of the device according to the present invention the exit opening of the nozzle cap has a diameter of at least 0.5 mm, preferably at least 1 mm, more preferably at least 1.5 mm, and even more preferably at least 2 mm. In an embodiment of the device according to the present invention the exit opening of the nozzle cap has a diameter of at most 9 mm, preferably at most 8 mm, more preferably at most 7 mm, and even more preferably at most 6 mm.

(32) In an embodiment of the device according to the present invention the exit opening of the nozzle cap is adjustable in size. The exit opening may for example be adjustable in size by means of an iris diaphragm, such as used in a photo camera.

(33) This embodiment offers the advantage that the size of the exit opening of the nozzle cap can easily be adapted for the use of the device according to the present invention in different applications, without having to replace the nozzle cap or making use of different devices according to the present invention having nozzle caps with exit openings of different sizes.

(34) The present invention further provides a method for generating a plasma jet. Preferably, the plasma jet comprises a monomer precursor. Preferably; the plasma jet is for plasma polymerisation or plasma activation, but not necessarily limited thereto. The method comprises the step of flowing a first ignitable gas through an inner channel towards an exit opening of the inner channel. The

method comprises the step of generating a plasma from the first ignitable gas flowing in the inner channel by means of a high voltage electrode arranged in the inner channel. The method comprises the step of flowing a monomer precursor through a middle channel towards an exit opening of the middle channel. The middle channel surrounds the inner channel. The middle channel is electrically insulated from the inner channel. The method comprises the step of flowing a second ignitable gas through an outer channel towards an exit opening of the outer channel. The outer channel surrounds the middle channel. The outer channel is electrically insulated from the middle channel. The method comprises the step of mixing the plasma of the first ignitable gas, the monomer precursor and the second ignitable gas in a mixing volume of a nozzle cap located after the exit openings of the inner channel, the middle channel and the outer channel. The method comprises the step of ejecting the mixture of the plasma of the first ignitable gas, the monomer precursor and the second ignitable gas in the mixing volume out of an exit opening of the nozzle cap located after the mixing volume. The exit opening of the nozzle cap is smaller than the exit opening of the outer channel.

(35) In an embodiment of the method according to the present invention the first ignitable gas is flowed at a predetermined first flow rate through the inner channel towards the exit opening of the inner channel. In an embodiment of the method according to the present invention the first flow rate is at least 0.1 slm, preferably at least 0.2 slm, more preferably at least 0.3 slm, and even more preferably at least 0.4 slm. In an embodiment of the method according to the present invention the first flow rate is at most 0.9 slm, preferably at most 0.8 slm, more preferably at most 0.7 slm, and even more preferably at most 0.6 slm. In an embodiment of the method according to the present invention the first flow rate is most preferably 0.5 slm.

(36) In an embodiment of the method according to the present invention the monomer precursor is flowed at a predetermined second flow rate through the middle channel towards the exit opening of the middle channel. In an embodiment of the method according to the present invention the second flow rate is at least 50 sccm, preferably at least 100 sccm, more preferably at least 150 sccm, and even more preferably at least 200 sccm. In an embodiment of the method according to the present invention the second flow rate is at most 650 sccm, preferably at most 550 sccm, more preferably at most 450 sccm, and even more preferably at most 350 sccm. In an embodiment of the method according to the present invention the second flow rate is most preferably 250 sccm.

(37) In an embodiment of the method according to the present invention the second ignitable gas is flowed at a predetermined third flow rate through the outer channel towards the exit opening of the outer channel. In an embodiment of the method according to the present invention the third flow rate is at least 0.2 slm, preferably at least 0.4 slm, more preferably at least 0.6 slm, and even more preferably at least 0.8 slm. In an embodiment of the method according to the present invention the third flow rate is at most 5 slm, preferably at most 4 slm, more preferably at most 3 slm, and even more preferably at most 2 slm. In an embodiment of the method according to the present invention the third flow rate is most preferably 1 slm.

(38) In an embodiment of the method according to the present invention the method is performed by means of the device according to the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be further elucidated by means of the following description and the appended figures.

(2) FIG. 1A shows a cutaway view of a nozzle of a device according to an embodiment of the present invention.

(3) FIG. 1B shows a cross section through the nozzle of FIG. 1A.

(4) FIG. 2A shows Schlieren images of a plasma jet generated by a device according to an

embodiment of the present invention with only the inner channel activated (left), with the inner and middle channel activated (middle), and with the inner, middle and outer channel activated (right) of the nozzle.

(5) FIG. 2B shows Schlieren images of the effluent out of a device according to an embodiment of the present invention with only the inner and middle channel activated, first without the high voltage electrode activated and then with the high voltage electrode activated (two left most images), and with the inner, middle and outer channel activated, first without the high voltage electrode activated and then with the high voltage electrode activated (two right most images).

(6) FIG. 3A shows a bar chart of the temperature of a plasma jet generated by a device according to an embodiment of the present invention with only the inner channel activated, and for different flow rates in the inner channel.

(7) FIG. 3B shows a bar chart of the temperature of a plasma jet generated by a device according to an embodiment of the present invention with the inner and middle channel activated, and for different flow rates in the middle channel.

(8) FIG. 3C shows a bar chart of the temperature of a plasma jet generated by a device according to an embodiment of the present invention with inner, middle and outer channel activated, and for different flow rates in the outer channel.

(9) FIG. 4 shows a graph of the surface temperature of a substrate exposed to a plasma jet generated by means of a device according to an embodiment of the present invention with the inner and middle channel activated for different flow rates in the middle channel, and with the inner, middle and outer channel activated for different flow rates in the outer channel.

(10) FIG. 5 shows a picture of a plasma polymerised coating on a substrate applied by means of a device according to an embodiment of the present invention with the inner and middle channel activated (right), and with the inner, middle and outer channel activated (left).

(11) FIG. 6A shows a graph of the atomic concentration of a substrate detectable through a plasma polymerised coating applied by means of a device according to an embodiment of the present invention with the outer channel activated (ON) and with the outer channel not activated (OFF).

(12) FIG. 6B shows the same graph as FIG. 6A, but after the substrate with the plasma polymerised coating has been submerged for 24 hours underwater.

MODES FOR CARRYING OUT THE INVENTION

(13) The present invention will be described with respect to particular embodiments and with reference to certain drawings but the invention is not limited thereto but only by the claims. The drawings described are only schematic and are non-limiting. In the drawings, the size of some of the elements may be exaggerated and not drawn on scale for illustrative purposes. The dimensions and the relative dimensions do not necessarily correspond to actual reductions to practice of the invention.

(14) Furthermore, the terms first, second, third and the like in the description and in the claims, are used for distinguishing between similar elements and not necessarily for describing a sequential or chronological order. The terms are interchangeable under appropriate circumstances and the embodiments of the invention can operate in other sequences than described or illustrated herein.

(15) Moreover, the terms top, bottom, over, under and the like in the description and the claims are used for descriptive purposes and not necessarily for describing relative positions. The terms so used are interchangeable under appropriate circumstances and the embodiments of the invention described herein can operate in other orientations than described or illustrated herein.

(16) The term “comprising”, used in the claims, should not be interpreted as being restricted to the means listed thereafter; it does not exclude other elements or steps. It needs to be interpreted as specifying the presence of the stated features, integers, steps or components as referred to, but does not preclude the presence or addition of one or more other features, integers, steps or components, or groups thereof. Thus, the scope of the expression “a device comprising means A and B” should not be limited to devices consisting only of components A and B. It means that with respect to the

present invention, the only relevant components of the device are A and B.

(17) Within the context of the present invention, the term “standard litre per minute”, slm, is a unit of volumetric flow rate of a fluid in “litre per minute”, l/min, at standard conditions for temperature and pressure, STP, which is defined as a temperature of 273.15 K and an absolute pressure of 101.325 kPa.

(18) Within the context of the present invention, the term “standard cubic centimetre per minute”, sccm, is a unit of volumetric flow rate of a fluid in “cubic centimetre per minute”, cm³/min, at standard conditions for temperature and pressure, STP, which is defined as a temperature of 273.15 K and an absolute pressure of 101.325 kPa.

(19) Within the context of the present invention, the term “ignitable gas” should be understood as a gas that is suitable for generating a plasma therefrom.

(20) FIGS. 1A and 1B show a nozzle **100** of a device according to the present invention for generating a plasma jet. The nozzle **100** extends along a longitudinal direction L.

(21) The nozzle **100** comprises an inner channel **110**. The inner channel **110** is formed in a first quartz capillary **112** that is arranged along the longitudinal direction L in the nozzle **100**. The device is configured for flowing a first ignitable gas through the inner channel **110** towards an exit opening **114** of the inner channel **110**. The first ignitable gas is a gas that is suitable for generating a plasma therefrom, such as for example He, Ar, N₂ and O₂, but any other suitable gas can be used. The inner channel **110** has a circular cross section and a circular exit opening **114**, but any other suitable shape may be used. Within the first quartz capillary **112**, and thus in the inner channel **110**, there is arranged a high voltage electrode **140** for generating a plasma from the first ignitable gas flowing in the inner channel **110**. The high voltage electrode **140** is a rod with a circular cross section, but any other suitable shape may be used. Preferably, the high voltage electrode **140** is used for Radio Frequency, RF, plasma generation of the first ignitable gas. The high voltage electrode **140** is needle shaped and extends along the longitudinal direction L in the inner channel **110** up to the exit opening **114** of the inner channel **110**, where the high voltage electrode **140** has a pointed tip **142** for an increased electrical field strength.

(22) The nozzle **100** also comprises a middle channel **120**. The middle channel **120** is formed between a second quartz capillary **122** and the first quartz capillary **112**, wherein the second quartz capillary **122** is arranged along the longitudinal direction L in the nozzle **100** and surrounds the first quartz capillary **112**. The device is configured for flowing a monomer precursor through the middle channel **120** towards and exit opening **124** of the middle channel **120**. The monomer precursor may be comprised in a flow of an ignitable gas. The monomer precursor may be for example styrene or methyl methacrylate, MMA, but any other suitable monomer precursor may be used depending on the application for the plasma jet, for example for plasma polymerisation or plasma activation. The middle channel **120** has a circular cross section and a circular exit opening **124**, but any other suitable shape may be used, preferably corresponding to the shape of the inner channel **110** and its exit opening **114**. The middle channel **120** is electrically insulated from the inner channel **110** by means of the first quartz capillary **112**, to prevent plasma generation of an ignitable gas in the middle channel **120** and deposition of polymerised monomer precursor in the middle channel **120**.

(23) The nozzle **100** also comprises an outer channel **130**. The outer channel **130** is formed between an outer wall **103**, **107** of the nozzle **100** and the second quartz capillary **122**, wherein the outer wall **103**, **107** of the nozzle **100** extends along the longitudinal direction L and surrounds the second quartz capillary **122**. The device is configured for flowing a second ignitable gas through the outer channel **130** towards an exit opening **134** of the outer channel **134**. The second ignitable gas may be the same gas as the first ignitable gas, but the second ignitable gas can also be a gas different from the first ignitable gas. The outer channel **130** has a circular cross section and a circular exit opening **134**, but any other suitable shape may be used, preferably corresponding to the shape of the inner channel **110** and the middle channel **120** and their respective exit openings **114**, **124**. The outer channel **130** is electrically insulated from the middle channel **120** by means of

the second quartz capillary **122**, to prevent plasma generation of the second ignitable gas in the outer channel **130**.

(24) The first quartz capillary **112**, the second quartz capillary **122** and the outer wall **103**, **107** of the nozzle **100** are coaxial, such that the inner channel **110**, the middle channel **120** and the outer channel **130** are coaxial, and such that the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130**, which are located in the same plane, are concentric.

(25) The nozzle **100** also comprises a nozzle cap **150**. The nozzle cap **150** extends along the longitudinal direction L. An upper portion **156** of the nozzle cap **150** forms a part of the outer wall **107** of the nozzle **100** that surrounds the outer channel **130** of the nozzle **100**. A lower portion **158** of the nozzle cap **150** extends beyond the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130**, and encloses there a mixing volume **152**. The mixing volume **152** is located within the nozzle cap **150**. At the end of the mixing volume **152** the nozzle cap **150** is provided with an exit opening **154**, which is also the exit opening **154** of the nozzle **100**. The mixing volume **152** is located directly after the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130**, and the mixing volume **152** extends from the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130** to the exit opening **154** of the nozzle cap **150**. The exit opening **154** is a circular opening, but any other suitable shape may be used, preferably a shape corresponding to the shape of the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130**. The nozzle cap **150** is coaxial with the outer channel **130**, the middle channel **120** and the inner channel **110**, such that the exit opening **154** of the nozzle cap **154** is coaxial with the exit openings **114**, **124**, **134** of the inner channel **110**, the middle channel **120** and the outer channel **130**.

(26) In the embodiment shown, the exit opening **154** of the nozzle cap **150** has a size which is smaller than the size of the exit opening **124** of the middle channel **120**, i.e. the exit opening **154** has a circumference or outer perimeter which is smaller than, i.e. falls within, the circumference or outer perimeter of the exit opening **124** of the middle channel **120**. In alternative embodiments, the size of the exit opening **154** of the nozzle cap **150** can however be smaller or larger, but is at least smaller than the size of the exit opening **134** of the outer channel **130**. Preferably, the exit opening **154** of the nozzle cap **150** has a diameter of at least 0.5 mm, preferably at least 1 mm, more preferably at least 1.5 mm, even more preferably at least 2 mm, and at most 9 mm, preferably at most 8 mm, more preferably at most 7 mm, even more preferably at most 6 mm.

(27) The inner wall **153** of the lower portion **158** of the nozzle cap **150** which encloses the mixing volume **152** is funnel shaped for gradually guiding, within the mixing volume **152**, the flow of the second ignitable gas that is coming out of the exit opening **134** of the outer channel **130** towards the exit opening **154** of the nozzle cap **150**, and thus directed towards the flow of the monomer precursor coming out of the exit opening **124** of the middle channel **120** and the flow of the plasma and possible leftover first ignitable gas coming out of the exit opening **114** of the inner channel **110**.

(28) The exit opening **154** of the nozzle cap **150**, in this embodiment, being smaller than the exit opening **124** of the middle channel **120**, and the flow of the second ignitable gas being directed within the mixing volume **152** towards the exit opening **154** of the nozzle cap **150** has the effect that the flows of the plasma and possible leftover first ignitable gas out of the inner channel **110**, the monomer precursor out of the middle channel **120** and the second ignitable gas out of the outer channel **130** do not directly exit the mixing volume **152** through the exit opening **154** of the nozzle cap **150** but are circulated for a while inside the mixing volume **152**. With these flows being circulated inside the mixing volume **152** of the nozzle cap **150** a better mixing of the monomer precursor with the plasma and the ignitable gases inside the mixing volume **152** is achieved, and thus in the plasma jet that is ejected out of the exit opening **154** of the nozzle cap **150**. This results in a more complete polymerisation of the monomer precursor in the plasma jet.

(29) In a preferred embodiment, to achieve an optimal mixing of the monomer precursor with the plasma and the ignitable gases inside the mixing volume **152** of the nozzle cap **150**, a first flow rate of the first ignitable gas in the inner channel **110** is at least 0.1 slm, preferably at least 0.2 slm, more preferably at least 0.3 slm, even more preferably at least 0.4 slm, and at most 0.9 slm, preferably at most 0.8 slm, more preferably at most 0.7 slm, even more preferably at most 0.6 slm, and most preferably 0.5 slm. A second flow rate of the monomer precursor in the middle channel **120** is at least 50 sccm, preferably at least 100 sccm, more preferably at least 150 sccm, even more preferably at least 200 sccm, and at most 650 sccm, preferably at most 550 sccm, more preferably at most 450 sccm, even more preferably at most 350 sccm, and most preferably 250 sccm. A third flow rate of the second ignitable gas in the outer channel **130** is at least 0.2 slm, preferably at least 0.4 slm, more preferably at least 0.6 slm, even more preferably at least 0.8 slm, and/or at most 5 slm, preferably at most 4 slm, more preferably at most 3 slm, even more preferably at most 2 slm, and most preferably 1 slm.

(30) The nozzle cap **150** is removably connected to the nozzle **100**, which is beneficial for cleaning the nozzle cap **150** or for replacing the nozzle cap **150** with a different type of nozzle cap **150**, for example a nozzle cap **150** having a differently sized exit opening **154**. For removably connecting the nozzle cap **150** to the nozzle **100**, the outer wall **103** at an upper portion **102** of the nozzle **100** is provided with an external screw thread **104** and the upper portion **156** of the nozzle cap **150** is provided with an internal screw thread **157**, such that the nozzle cap **150** can be screwed onto the outer wall **103** of the upper portion **102** of the nozzle **100**. It should however be clear, that the nozzle **100** and the nozzle cap **150** may also be provided with different means for removably connecting the nozzle cap **150** to the nozzle **100**.

(31) The flow of the first ignitable gas out of the inner channel **110**, the monomer precursor out of the middle channel **120** and the second ignitable gas out of the outer channel **130** being circulated in the mixing volume **152** of the nozzle **100**, for a more uniform distribution of the monomer precursor in the plasma jet, is demonstrated experimentally as further detailed below.

(32) FIG. 2A shows Schlieren images of a plasma jet generated by means of a device according to an embodiment of the present invention. For this experiment a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. In the left image, only the inner channel **110** was active with a flow rate of argon gas, Ar, of 0.5 slm. For the middle image, the middle channel **120** was also activated, in addition to the inner channel **110**, with a flow rate of Ar of 250 sccm. Here, for demonstration purposes Ar was chosen, although it is not a monomer precursor. For the right image, the outer channel **130** was also activated, in addition to the inner channel **110** and the middle channel **120**, with a flow rate of Ar of 2 slm.

(33) FIG. 2A shows that when only the inner channel **110** is in use, as shown in the left image, the plasma jet has the longest section of laminar flow regime. When the inner channel **110** and the middle channel **120** are in use, as shown in the middle image, the length of the section of laminar flow regime decreases due to the higher total flow out of the exit opening **154** of the nozzle **100**. When the inner channel **110**, the middle channel **120** and the outer channel **130** are all in use, the transition to turbulent regime happens at a closer distance from the exit opening **154** of the nozzle **100** as even higher total flow is passing through said exit opening **154**. Moreover, in this case, the plasma jet receives additional thrust due to the mixing volume **152** in the nozzle cap **150** where the flows coming out of the inner channel **110**, the middle channel **120** and the outer channel **130** circulate for a while.

(34) FIG. 2B shows Schlieren images of an effluent coming out of a device according to an embodiment of the present invention. For this experiment a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. In the left two images, the inner channel **110** was active with a flow rate of Ar of 0.5 slm and the middle channel **120** with a flow rate of Ar and styrene of 250 sccm. In the first of these two images, the high voltage electrode **140** of the device was not active, such that the effluent is merely the

combination of the flows out of the inner channel **110** and the middle channel **120**. In the second of these two images, the high voltage electrode **140** of the device was active, such that the effluent is a plasma jet. For the right two images, the outer channel **130** was also activated, in addition to the inner channel **110** and the middle channel **120**, with a flow rate of Ar of 2 slm. In the first of these two images, the high voltage electrode **140** of the device was not active, such that the effluent is merely the combination of the flows out of the inner channel **110**, the middle channel **120** and the outer channel **130**. In the second of these two images, the high voltage electrode **140** of the device was active, such that the effluent is a plasma jet.

(35) FIG. 2B shows that when only the inner channel **110** and middle channel **120** are in use, the effluent flow transits to the turbulent regime close to the exit opening **154** of the nozzle **100**. When the inner channel **110**, the middle channel **120** and the outer channel **130** are all in use, additional mixing occurs inside the mixing volume **152** of the nozzle cap **150**. This extra volume provides time and space to form a “united” stream that flows out of the exit opening **154** of the nozzle **100**. A higher gas temperature, caused by the flows coming out of the inner channel **110**, the middle channel **120** and the outer channel **130** being circulated inside the mixing volume **152** and being exposed to the pointed tip **142** of the high voltage electrode **140** and thus heated thereby, may also explain the better contrast in the second image of the two right images in comparison to the second image of the two left images.

(36) The aforementioned increase in temperature of the plasma jet due to the flows coming out of the inner channel **110**, the middle channel **120** and the outer channel **130** being circulated inside the mixing volume **152** of the nozzle **100**, is also demonstrated by means of the experiments of FIGS. 3A-C and the experiment of FIG. 4.

(37) FIG. 3A-C show the temperature, i.e. the gas temperature TGAS, the rotational temperature estimated for OH, Trot OH, and the rotational temperature estimated for N2, Trot N2, of the plasma jet coming out of the exit opening **154** of the nozzle **100**, measured by means of Rayleigh scattering of a laser beam pointed through the plasma jet. For these experiments a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. For the first experiment of FIG. 3A, only the inner channel **110** was active with flow rates of Ar of 0.5 slm, 1 slm and 1.5 slm. For the second experiment of FIG. 3B, the middle channel **120** was also activated with flow rates of Ar of 0 sccm, 50 sccm, 250 sccm and 500 sccm, and this in addition to the inner channel **110** having a flow rate of Ar of 0.5 slm. Here, for demonstration purposes Ar was used in the middle channel **120**, although it is not a monomer precursor. For the third experiment of FIG. 3C, the outer channel **130** was also activated with flow rates of Ar of 0 slm, 1 slm, 2 slm and 3 slm, and this in addition to the inner channel **110** having a flow rate of Ar of 0.5 slm and the middle channel **120** having a flow rate of Ar of 250 sccm.

(38) FIGS. 3A-C show that increasing the flow rate in any one of the inner channel **110**, the middle channel **120** and the outer channel **130** causes a decrease in temperature of the plasma jet. As can be seen in FIG. 3B, adding Ar gas to the middle channel **120**, while keeping a 0.5 slm flow rate of Ar in the inner channel **110**, does not change the temperature significantly due to small flow rates. As can be seen in FIG. 3C, the highest temperatures are obtained when the inner channel **110**, the middle channel **120** and the outer channel **130** are all in use. This outcome is explained by a prolonged stay of the flows coming out of the inner channel **110**, the middle channel **120** and the outer channel **130** inside the mixing volume **152** of the nozzle cap **150**, due to said flows being circulated for a while inside the mixing volume **152**.

(39) FIG. 4 shows the surface temperature of a substrate exposed to a plasma jet generated by means of a device according to an embodiment of the present invention, wherein the surface temperature was measured by means of an infrared camera and at the centre of the exposed area. For this experiment a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. For a first series of measurements, the inner channel **110** was active with an Ar flow rate of 0.5 slm and the middle channel **120** with flow rates

of Ar and styrene of 50 sccm (.square-solid.), 250 sccm (.circle-solid.) and 500 sccm (.box-tangle-solidup.). For a second series of measurements, the outer channel **130** was also activated with flow rates of Ar of 1 slm (.Math.), 2 slm) (.diamond-solid.) and 3 slm () custom character), and this in addition to the inner channel **110** having a flow rate of Ar of 0.5 slm and the middle channel **120** having a flow rate of Ar and styrene of 250 sccm.

(40) FIG. **4** shows that increasing the flow rates results in a lower temperature of the substrate at all experimental conditions. The flow rate of Ar and styrene in the middle channel does not significantly influence the substrate temperature. Significantly higher substrate temperatures were observed when the inner channel **110**, the middle channel **120** and the outer channel **130** were all in use. This is due to the gas dynamic inside the mixing volume **152** of the nozzle cap **150**, which causes the flows coming out of the inner channel **110**, the middle channel **120** and the outer channel **130** being exposed to the pointed tip **142** of the high voltage electrode **140** for longer and thus heated thereby.

(41) The uniform distribution of the monomer precursor in the plasma jet is demonstrated by means of the experiment of FIG. **5** and the experiments of FIGS. **6A** and **6B**.

(42) FIG. **5** shows a picture of a plasma polymerised coating on a silicon substrate applied by means of a device according to an embodiment of the present invention. For this experiment a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. The coating was deposited on the substrate by moving the plasma jet backwards and forwards along a straight line above the substrate. The deposition speed was 2 mm/s, the deposition duration was 5 min and the moving distance was 30 mm. For a first experiment, of which the results are shown on the right, the inner channel **110** was active with a flow rate of Ar of 0.5 slm and the middle channel **120** with a flow rate of Ar and styrene of 250 sccm. For a second experiment, of which the results are shown on the left, the outer channel **130** was also activated with a flow rate of Ar of 2 slm, and this in addition to the inner channel **110** having a flow rate of Ar of 0.5 slm and the middle channel **120** having a flow rate of Ar and styrene of 250 sccm.

(43) FIG. **5** shows that the coating deposited without employing the outer channel **130**, as can be seen on the right, is narrow due to the styrene monomer precursor in the plasma jet being located close to the centre of the plasma jet. The coating deposited with the inner channel **110**, the middle channel **120** and the outer channel **130** all active, as can be seen on the left, is more spread out and thus wider, due to the styrene monomer precursor being more uniformly distributed in the plasma jet.

(44) FIG. **6A** shows the atomic concentration of a silicon substrate, i.e. the surface silicon concentration of the silicon substrate, detectable through a plasma polymerised coating applied by means of a device according to an embodiment of the present invention, wherein the atomic concentration was measured by means of X-ray photoelectron spectroscopy, XPS, along the width of the coating. For this experiment a device was used with a nozzle **100** of which the mixing volume **152** has a circular exit opening **154** with a diameter of 4.5 mm. The coating was deposited on the substrate by moving the plasma jet backwards and forwards along a straight line above the substrate. The deposition speed was 2 mm/s, the deposition duration was 5 min and the moving distance was 30 mm. For a first measurement (.square-solid.), the inner channel **110** was active with a flow rate of Ar of 0.5 slm, the middle channel **120** with a flow rate of Ar and styrene of 250 sccm, and the outer channel **130** was not active (OFF). For a second measurement (.circle-solid.), the outer channel **130** was also activated (ON) with a flow rate of Ar of 2 slm, and this in addition to the inner channel **110** having a flow rate of Ar of 0.5 slm and the middle channel **120** having a flow rate of Ar and styrene of 250 sccm.

(45) FIG. **6A** shows that the coating deposited with the inner channel **110**, the middle channel **120** and the outer channel **130** all active, fully covers the silicon substrate over the entire width of the coating, due to the styrene monomer precursor being uniformly distributed in the plasma jet. The

coating deposited without employing the outer channel **130**, on the other hand, only fully covers the silicon substrate in a smaller range, due to the styrene monomer precursor in the plasma jet being located close to the centre of the plasma jet.

(46) FIG. **6B** shows for the coated silicon substrate of the experiment of FIG. **6A**, the atomic concentration of the silicon substrate, i.e. the surface silicon concentration of the silicon substrate, detectable through the plasma polymerised coating, but now after the coated silicon substrate has been submerged in water for a period of 24 hours.

(47) FIG. **6B** shows that the coating deposited with the inner channel **110**, the middle channel **120** and the outer channel **130** all active, is more stable after having been submerged in water in comparison to the coating deposited without employing the outer channel **130**, which is a further advantage of the device according to the present invention.

(48) TABLE-US-00001 References 100 nozzle 102 upper portion 103 outer wall 104 external screw thread 106 lower portion 107 outer wall 110 inner channel 112 first quartz capillary 114 exit opening 120 middle channel 122 second quartz capillary 124 exit opening 130 outer channel 134 exit opening 140 high voltage electrode 142 pointed tip 150 nozzle cap 152 mixing volume 153 inner wall 154 exit opening 156 upper portion 157 internal screw thread 158 lower portion L longitudinal direction

Claims

1. A device for generating a plasma jet, wherein a nozzle of the device comprises: an inner channel, wherein the device is configured for flowing a first ignitable gas through the inner channel towards an exit opening of the inner channel, wherein a high voltage electrode is arranged in the inner channel for generating a plasma from the first ignitable gas flowing in the inner channel; a middle channel, wherein the middle channel surrounds the inner channel, wherein the middle channel is electrically insulated from the inner channel, wherein the device is configured for flowing a monomer precursor through the middle channel towards an exit opening of the middle channel; an outer channel, wherein the outer channel surrounds the middle channel, wherein the outer channel is electrically insulated from the middle channel, wherein the device is configured for flowing a second ignitable gas through the outer channel towards an exit opening of the outer channel; and a nozzle cap, wherein the nozzle cap encloses a mixing volume located after the exit openings of the inner channel, the middle channel and the outer channel, and wherein the nozzle cap is provided with an exit opening after the mixing volume, wherein the exit opening of the nozzle cap is smaller than the exit opening of the outer channel.
2. The device according to claim 1, wherein the mixing volume is adapted for mixing the plasma of the first ignitable gas from the inner channel, the monomer precursor from the middle channel and the second ignitable gas from the outer channel.
3. The device according to claim 1, wherein the mixing volume is located within the nozzle cap.
4. The device according to claim 1, wherein an inner wall of the nozzle cap encloses the mixing volume.
5. The device according to claim 1, wherein the mixing volume is located directly after the exit openings of the inner channel, the middle channel and the outer channel.
6. The device according to claim 1, wherein the exit opening of the nozzle cap is located directly after the mixing volume.
7. The device according to claim 1, wherein the mixing volume extends from the exit openings of the inner channel, the middle channel and the outer channel to the exit opening of the nozzle cap.
8. The device according to claim 1, wherein the exit opening of the nozzle cap is smaller than or equal to the exit opening of the middle channel.
9. The device according to claim 1, wherein an inner wall of the nozzle cap is shaped for gradually guiding a flow of the second ignitable gas in the mixing volume from the exit opening of the outer

channel towards the exit opening of the nozzle cap.

10. The device according to claim 9, wherein the inner wall of the nozzle cap is funnel shaped.

11. The device according to claim 1, wherein the nozzle cap is arranged for being removably connected to the nozzle of the device.

12. The device according to claim 1, wherein the device is configured for flowing the first ignitable gas at a predetermined first flow rate through the inner channel towards the exit opening of the inner channel, wherein a first flow rate is at least 0.1 slm, preferably at least 0.2 slm, more preferably at least 0.3 slm, even more preferably at least 0.4 slm, and/or at most 0.9 slm, preferably at most 0.8 slm, more preferably at most 0.7 slm, even more preferably at most 0.6 slm, and most preferably 0.5 slm.

13. The device according to claim 1, wherein the device is configured for flowing the monomer precursor at a predetermined second flow rate through the middle channel towards the exit opening of the middle channel, wherein a second flow rate is at least 50 sccm, preferably at least 100 sccm, more preferably at least 150 sccm, even more preferably at least 200 sccm, and/or at most 650 sccm, preferably at most 550 sccm, more preferably at most 450 sccm, even more preferably at most 350 sccm, and most preferably 250 sccm.

14. The device according to claim 1, wherein the device is configured for flowing a carbon based monomer precursor through the middle channel towards the exit opening of the middle channel.

15. The device according to claim 1, wherein the device is configured for flowing the second ignitable gas at a predetermined third flow rate through the outer channel towards the exit opening of the outer channel, wherein a third flow rate is at least 0.2 slm, preferably at least 0.4 slm, more preferably at least 0.6 slm, even more preferably at least 0.8 slm, and/or at most 5 slm, preferably at most 4 slm, more preferably at most 3 slm, even more preferably at most 2 slm, and most preferably 1 slm.

16. The device according to claim 1, wherein the exit opening of the nozzle cap has a diameter of at least 0.5 mm, preferably at least 1 mm, more preferably at least 1.5 mm, even more preferably at least 2 mm, and/or at most 9 mm, preferably at most 8 mm, more preferably at most 7 mm, even more preferably at most 6 mm.

17. The device according to claim 1, wherein the exit opening of the nozzle cap is adjustable in size.

18. A method for generating a plasma jet, wherein the method comprises the steps of: flowing a first ignitable gas through an inner channel towards an exit opening of the inner channel, and generating a plasma from the first ignitable gas flowing in the inner channel by a high voltage electrode arranged in the inner channel; flowing a monomer precursor through a middle channel towards an exit opening of the middle channel, wherein the middle channel surrounds the inner channel, and wherein the middle channel is electrically insulated from the inner channel; and flowing a second ignitable gas through an outer channel towards an exit opening of the outer channel, wherein the outer channel surrounds the middle channel, wherein the outer channel is electrically insulated from the middle channel; mixing the plasma of the first ignitable gas, the monomer precursor and the second ignitable gas in a mixing volume of a nozzle cap located after the exit openings of the inner channel, the middle channel and the outer channel; and ejecting the mixture of the plasma of the first ignitable gas, the monomer precursor and the second ignitable gas in the mixing volume out of an exit opening of the nozzle cap located after the mixing volume, wherein the exit opening of the nozzle cap is smaller than the exit opening of the outer channel.

19. The method according to claim 18, wherein the first ignitable gas is flowed at a predetermined first flow rate through the inner channel towards the exit opening of the inner channel, wherein a first flow rate is at least 0.1 slm, preferably at least 0.2 slm, more preferably at least 0.3 slm, even more preferably at least 0.4 slm, and/or at most 0.9 slm, preferably at most 0.8 slm, more preferably at most 0.7 slm, even more preferably at most 0.6 slm, and most preferably 0.5 slm.

20. The method according to claim 18, wherein the monomer precursor is flowed at a

predetermined second flow rate through the middle channel towards the exit opening of the middle channel, wherein a second flow rate is at least 50 sccm, preferably at least 100 sccm, more preferably at least 150 sccm, even more preferably at least 200 sccm, and/or at most 650 sccm, preferably at most 550 sccm, more preferably at most 450 sccm, even more preferably at most 350 sccm, and most preferably 250 sccm.

21. The method according to claim 18, wherein the second ignitable gas is flowed at a predetermined third flow rate through the outer channel towards the exit opening of the outer channel, wherein a third flow rate is at least 0.2 slm, preferably at least 0.4 slm, more preferably at least 0.6 slm, even more preferably at least 0.8 slm, and/or at most 5 slm, preferably at most 4 slm, more preferably at most 3 slm, even more preferably at most 2 slm, and most preferably 1 slm.

22. The method according to claim 18, wherein the method is performed by a device for generating a plasma jet, wherein a nozzle of the device comprises: an inner channel, wherein the device is configured for flowing a first ignitable gas through the inner channel towards an exit opening of the inner channel, wherein a high voltage electrode is arranged in the inner channel for generating a plasma from the first ignitable gas flowing in the inner channel; a middle channel, wherein the middle channel surrounds the inner channel, wherein the middle channel is electrically insulated from the inner channel, wherein the device is configured for flowing a monomer precursor through the middle channel towards an exit opening of the middle channel; an outer channel, wherein the outer channel surrounds the middle channel, wherein the outer channel is electrically insulated from the middle channel, wherein the device is configured for flowing a second ignitable gas through the outer channel towards an exit opening of the outer channel; and a nozzle cap, wherein the nozzle cap encloses a mixing volume located after the exit openings of the inner channel, the middle channel and the outer channel, and wherein the nozzle cap is provided with an exit opening after the mixing volume, wherein the exit opening of the nozzle cap is smaller than the exit opening of the outer channel.
